ReceiptLedger

Web Operations Console & Desktop Dashboard Design Brief

Monorepo: web/ (React 18 + Vite)
API Gateway: http://localhost:8000/api/v1
Scope: Admin, Audit & Triage
Form Factor: Desktop (1280px+)
Design Constraint: Zero Gradients | Zero Glass

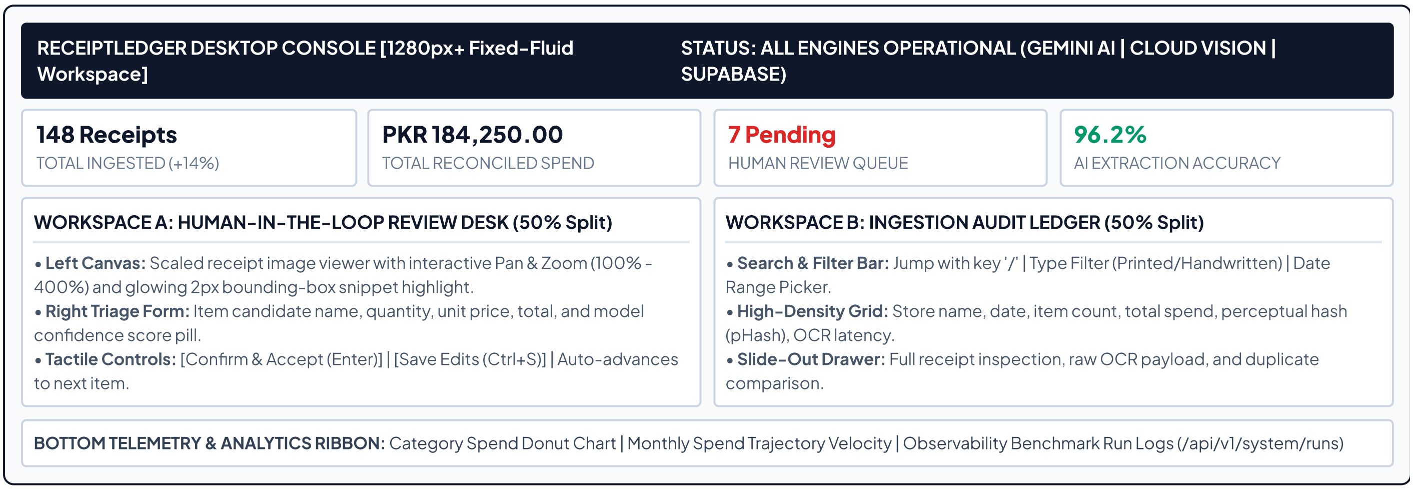
1. System Role & Architecture Overview

The Web Dashboard is the **central desktop operations, audit, and financial intelligence console** for ReceiptLedger. While the mobile client handles quick on-the-go receipt photography, the desktop console provides high-detail verification, audit logging, and observability for household finances or business bookkeeping.

Dimension	Mobile App (Flutter)	Web Dashboard (React / Vite)
Primary Role	Shopper / In-the-field Consumer	Operator / Household Accountant / Administrator
Key Workflow	Quick camera photo snap, crop, and upload	Triage desk, spending analytics, deep audit ledger
Interaction Mode	One-thumb touch on smartphone	Multi-column table, keyboard shortcuts, mouse hover
Data Density	Minimal (Automated capture)	High Detail (Raw OCR verification, hashes, latency)

2. Desktop Viewport & Spatial Layout Blueprint

The desktop dashboard is optimized for 1280px to 1920px+ viewports without unnecessary modal popups or hidden dropdowns. Below is the spatial architecture:



RECEIPTLEDGER DESKTOP CONSOLE [1280px+ Fixed-Fluid Workspace]

STATUS: ALL ENGINES OPERATIONAL (GEMINI AI | CLOUD VISION | SUPABASE)

148 Receipts
TOTAL INGESTED (+14%)

PKR 184,250.00
TOTAL RECONCILED SPEND

7 Pending
HUMAN REVIEW QUEUE

96.2%
AI EXTRACTION ACCURACY

WORKSPACE A: HUMAN-IN-THE-LOOP REVIEW DESK (50% Split)

- **Left Canvas:** Scaled receipt image viewer with interactive Pan & Zoom (100% - 400%) and glowing 2px bounding-box snippet highlight.
- **Right Triage Form:** Item candidate name, quantity, unit price, total, and model confidence score pill.
- **Tactile Controls:** [Confirm & Accept (Enter)] | [Save Edits (Ctrl+S)] | Auto-advances to next item.

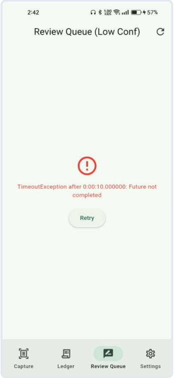
WORKSPACE B: INGESTION AUDIT LEDGER (50% Split)

- **Search & Filter Bar:** Jump with key '/' | Type Filter (Printed/Handwritten) | Date Range Picker.
- **High-Density Grid:** Store name, date, item count, total spend, perceptual hash (pHash), OCR latency.
- **Slide-Out Drawer:** Full receipt inspection, raw OCR payload, and duplicate comparison.

BOTTOM TELEMETRY & ANALYTICS RIBBON: Category Spend Donut Chart | Monthly Spend Trajectory Velocity | Observability Benchmark Run Logs (/api/v1/system/runs)

3. Self-Contained Visual References & Embedded Image Showcase

All relevant receipt samples and existing screen states are **embedded directly within this document**. No searching through repository folders required.

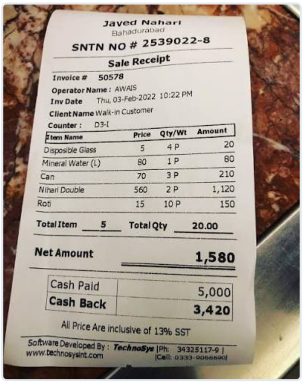


Review Queue (Low Conf)

TimeoutException after 0:00:10.000000: Future not completed

Reference 1: Current Mobile Review Screen

Shows existing barebones mobile review queue with raw technical exception. Replaced on desktop by our split-screen triage workspace.



Javed Nahari
Bahadurabad
SNTN NO # 2539022-8
Sale Receipt
Invoice # 50578
Operator Name : AWAIS
Inv Date Thu, 03-Feb-2022 10:22 PM
Client Name Walk-in Customer
Counter : D3-I

Item Name	Price	Qty/Wt	Amount
Disposible Glass	5	4 P	20
Mineral Water (L)	80	1 P	80
Can	70	3 P	210
Nihari Double	560	2 P	1,120
Roti	15	10 P	150

Total Item 5 Total Qty 20.00

Net Amount 1,580

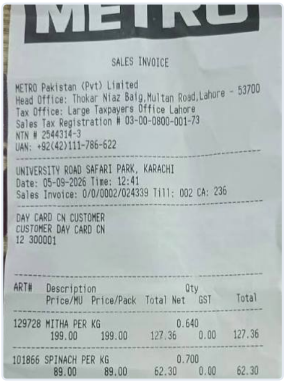
Cash Paid	5,000
Cash Back	3,420

All Price Are inclusive of 13% SST

Software Developed By : TechnoSys |Ph: 34325117-9 |
www.technosysint.com |Cell: 0333-9066690|

Reference 2: Real Pakistani POS Receipt

Javed Nihari POS receipt with bilingual Urdu/English headers, itemized dishes, GST sales tax lines, and thermal fade.



SALES INVOICE

METRO Pakistan (Pvt) Limited
Head Office: Thokar Niaz Baig,Multan Road,Lahore - 53700
Tax Office: Large Taxpayers Office Lahore
Sales Tax Registration # 03-00-0800-001-73
NTN # 2544314-3
UAN: +92(42)111-786-622

UNIVERSITY ROAD SAFARI PARK, KARACHI
Date: 05-09-2026 Time: 12:41
Sales Invoice: 0/0/0002/024339 Till: 002 CA: 236

DAY CARD CN CUSTOMER
CUSTOMER DAY CARD CN
12 300001

```
ART#   Description                   Qty
       Price/MU  Price/Pack  Total Net  GST     Total
129728 MITHA PER KG                 0.640
        199.00     199.00     127.36    0.00    127.36
101866 SPINACH PER KG               0.700
         89.00      89.00      62.30    0.00     62.30
```

Reference 3: Retail Superstore Receipt

Metro Cash & Carry receipt featuring high-density multi-column product codes, barcode numbers, and per-unit prices.

4. Receipt Ingestion, Deduplication & Verification Pipeline

```
[1. Upload Receipt Image] ──► [2. OpenCV Preprocessing & Deskew] ──► [3. Perceptual Hashing (Layer 1 & 2)] │
┌────────────────────────────────────────────────┴──────────────────────────────┐ ▼ ▼ Exact SHA256 OR pHash dist ≤ 4 Unique Receipt │ │ ▼ ▼ 409
Conflict Banner [4. Gemini AI Vision] (Link to existing receipt) │ ┌─────────────────────────┴─────────────────────────┐ ▼ ▼ Confidence ≥ 85%
Confidence < 85% │ │ ▼ ▼ status: PROCESSED status: PENDING_REVIEW (Direct to Ledger) (Added to Review Queue) │ ▼ [5. Desktop Review
Desk] 1-Click [Enter] / [Ctrl+S]
```

5. Desktop Keyboard Ergonomics & Shortcuts Cheat Sheet

Shortcut	Context	Action / Result
`Enter` / `Return`	Review Desk Table	1-Click Approve active item (`action: "ACCEPTED"`) and auto-focus next row
`Ctrl + S` / `Cmd + S`	Review Desk Table	Save inline edits on candidate name, qty, or price (`action: "EDITED"`)
`Tab`	Review Desk Row	Advance focus between candidate item name, quantity, unit, and price fields
`Shift + Tab`	Review Desk Row	Move backward to previous field
`↑` / `↓` or `J` / `K`	Audit Ledger Grid	Select previous / next receipt row
`Space`	Audit Ledger Row	Open Slide-Out Inspector Drawer for selected receipt
`Esc`	Modal / Drawer	Dismiss Slide-Out Inspector Drawer or cancel inline editing
/ (Forward Slash)	Global	Jump focus directly to Ingestion Ledger Search input

6. Backend API Directory & Concrete JSON Payloads

Method	Endpoint	Primary Purpose	Key Query / Body Params
`GET`	`/api/v1/health`	System health & engine quota	None (Returns Gemini & Cloud Vision status)
`GET`	`/api/v1/dashboard/overview`	Monthly spend, KPIs & category split	`year=2026&month=10`
`GET`	`/api/v1/review-queue`	Unresolved low-confidence items	`status_filter=UNRESOLVED`
`PATCH`	`/api/v1/review-queue/{id}`	1-click accept or inline edit	`confirmed_name`, `action`
`GET`	`/api/v1/analytics/receipts`	Searchable paginated audit ledger	`page=1&page_size=25&search=Imtiaz`
`GET`	`/api/v1/system/runs`	Observability & benchmark logs	None (Returns run IDs and accuracy %)

6.1 System Health (GET /api/v1/health)

```
{
  "status": "ok",
  "version": "1.0.0",
  "environment": "development",
  "engines": {
    "cloud_vision": "ok (quota_remaining=950)",
    "tesseract": "ok",
    "easyocr": "ok (lazy-loaded)",
    "gemini": "ok (model=gemini-3.5-flash-lite)",
    "database": "supabase"
  }
}
```

6.2 Review Queue Triage Item (GET /api/v1/review-queue)

```
[
  {
    "queue_id": "7a79f40f-3047-4d76-a577-a41a58af9a56",
    "line_item_id": "93b118b5-12ea-44d5-8bc2-a5d6f1406541",
    "receipt_id": "41fd8d2a-e322-4f5f-aa37-ed65ffab5519",
    "merchant_name": "Chaudhry Sabzi & Fruit Store",
    "raw_text": "129728 WITHA PER KG 0.640",
    "candidate_name": "Sweet Melon",
    "model_confidence": 0.745,
    "review_status": "UNRESOLVED"
  }
]
```

6.3 Resolving Item (PATCH /api/v1/review-queue/{queue_id})

```
// 1-Click Accept:
{ "confirmed_name": "Sweet Melon", "action": "ACCEPTED" }

// Inline Edit & Correct:
{ "confirmed_name": "Sweet Melon", "confirmed_quantity": 0.64, "confirmed_unit_price": 199.0, "action": "EDITED" }
```

7. Design Aesthetic, Constraints & Testing Controls

- **Rule #1: ZERO GRADIENTS.** No rainbow fills or shiny button gradients. Use solid colors with clean contrast.
- **Rule #2: ZERO GLASSMORPHISM.** No liquid acrylic blur (`backdrop-filter: blur`). Use clean 1px borders (`#CBD5E1`) and crisp white/slate cards.
- **Instant Test Controls:** Click `[Seed Demo Data]` (`POST /api/v1/dashboard/seed-demo`) to load 148 realistic receipts. Click `[Reset Ledger]` (`POST /api/v1/dashboard/reset`) to test zero-data states.
- **Default Admin Credentials:** `user@receiptledger.com` / `demo1234`.